

INTRODUCTION TO MACHINE LEARNING

REDUCTION WITH PCA AND
VISUALIZATION USING PCA AND UMAP
WORK 4

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1 Introduction

This study examines the effectiveness of Principal Component Analysis (PCA) for dimensionality reduction using two datasets from the UCI Machine Learning Repository. The analysis evaluates the impact of PCA on clustering performance by applying the K-Means++ and OPTICS algorithms to both the original datasets and their PCA-transformed versions. Additionally, the visualizations produced by PCA are compared to those generated by Uniform Manifold Approximation and Projection (UMAP) to assess their ability to preserve data structure and provide meaningful insights.

2 Datasets

2.1 Hepatitis

The Hepatitis dataset contains 155 instances and 19 attributes, including six numerical ones: patient age, bilirubin levels, alkaline phosphatase, SGOT levels, prothrombin time, and albumin levels. The remaining 13 attributes are categorical with binary values yes or no, covering clinical symptoms and treatment indicators such as gender, steroid use, antiviral treatment, and symptoms like fatigue, anorexia, and enlarged liver. The target class has two outcomes: DIE and LIVE. The dataset shows an imbalanced class distribution with 79.35% in the LIVE class (Maj.Cla.) and 20.65% in the DIE class, resulting in a class distribution deviation of 29.35%. It has a 6% of missing values across attributes excluding age, sex, antivirals, and histology.

2.2 Breast Cancer Wisconsin

The Breast Cancer Wisconsin dataset contains 699 instances, each characterized by 9 numeric attributes, with values ranging from 1 to 10. These attributes describe various cell characteristics, such as Clump Thickness, Cell Size Uniformity, and Mitoses. The target variable, *Class*, is categorical with two possible values: *benign* and *malignant*. The dataset’s majority class accounts for 70.28% of instances, while the minority class represents 29.72%, resulting in a class deviation of 20.28%. Additionally, it contains 0.25% missing values across its attributes.

3 Data preprocessing

Data preprocessing cleans, normalizes, and transforms raw data to ensure compatibility with machine learning algorithms, enhancing accuracy, reliability, and insights.

For the **Hepatitis dataset**, preprocessing was required due to mixed data types. Rows with over 30% missing values were removed, and numerical missing values were filled with the mean to maintain distribution. Categorical columns were decoded, mapped into binary form (e.g., “yes” to 1, “no” to 0), and remaining missing values were filled with the mode. Numerical columns were normalized using MinMaxScaler, scaling values between 0 and 1 for consistency and effective feature contribution.

For the **Breast Cancer Wisconsin dataset**, rows with over 30% missing values were removed, and remaining missing values in numeric columns were filled with the mean to preserve the overall distribution. All numerical attributes were normalized using MinMaxScaler, scaling their values between 0 and 1 to ensure consistency and balanced feature contribution. The target variable was decoded and mapped to binary values, with “benign” represented as 0 and “malignant” as 1.

Finally, the “class” column was dropped from both datasets to prepare for clustering, but it was saved separately for evaluating model performance later.

4 PCA

In this project, a Principal Component Analysis (PCA) algorithm was implemented from scratch to evaluate its effectiveness in dimensionality reduction. The process began by selecting two or three features for visualization, followed by computing the d-dimensional mean vector and the covariance matrix of the dataset to analyze its variability. Eigenvalues and their corresponding eigenvectors were then derived from the covariance matrix, sorted in descending order of significance. The top k eigenvectors were selected to form a transformation matrix, which was used to project the data into a lower-dimensional subspace. Lastly, to evaluate the quality of the dimensionality reduction, the reduced representation was reconstructed back to its original dimensionality, enabling a comparison between the original and reconstructed data.

4.1 Hepatitis

When applying PCA to the hepatitis dataset, the features *sgot*, *protime*, and *age* were selected for 2D and 3D representations. These specific features were chosen for being numerical, as they can be better visualized and analyzed compared to categorical features.

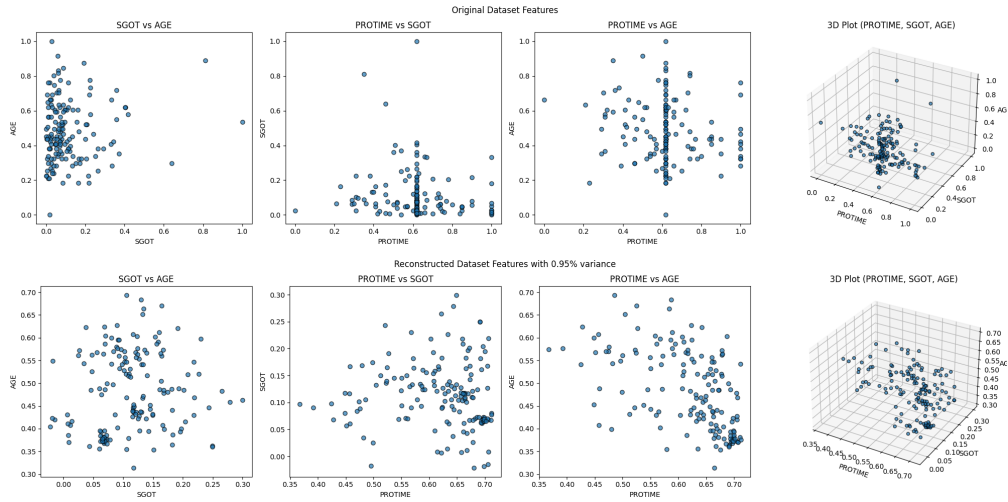


Figure 1: Pairwise and 3D plots of features **sgot**, **protime**, and **age** from the hepatitis dataset before and after applying PCA.

Figure 1 provides a visual comparison of the pairwise relationships and 3D scatter plots of the features *sgot*, *protime*, and *age* from the Hepatitis dataset before and after applying PCA. The upper row depicts the original dataset’s feature space, showcasing the raw relationships and distributions. The lower row illustrates the reconstructed features after dimensionality reduction with PCA, retaining 95% of the variance.

The visual comparison highlights PCA’s effectiveness in preserving the structure and patterns of the original data while reducing dimensionality. Pairwise scatter plots and 3D projections suggest key feature relationships remain consistent after PCA transformation. However, subtle changes in the spread and alignment of points reflect the variance captured in the principal components.

4.2 Breast Cancer

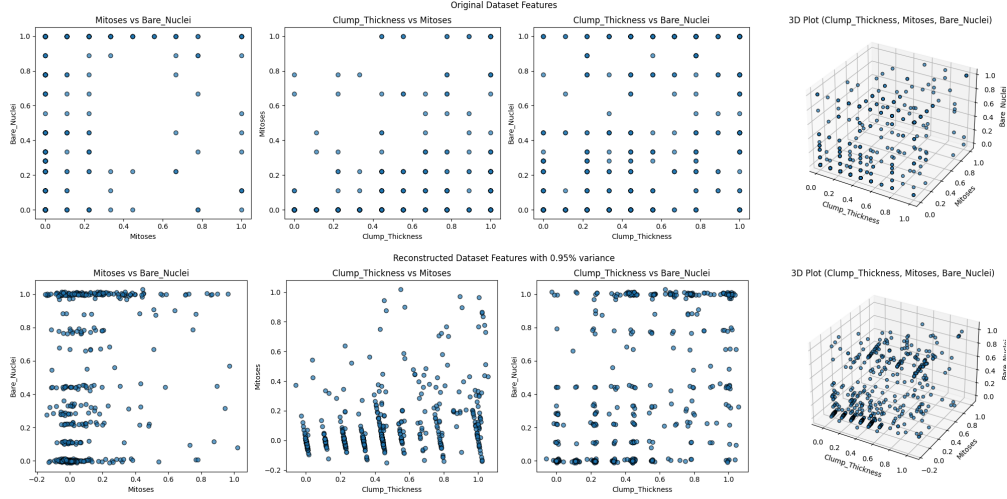


Figure 2: Pairwise and 3D plots of features **Mitoses**, **Bare_Nuclei**, and **Clump_Thickness** from the Brest Cancer dataset before and after applying PCA.

Figure 2 presents pairwise relationships and 3D scatter plots of the features *Mitoses*, *Bare_Nuclei*, and *Clump_Thickness* from the Breast Cancer dataset before and after applying PCA. Again, we selected these features over others because they are numerical. As in the Hepatitis dataset, the upper row illustrates the original feature space, showing the raw distributions and correlations between the selected features. The lower row displays the reconstructed features after PCA transformation, retaining 95% of the dataset’s variance.

Again, this comparison emphasizes how PCA effectively preserves the primary structure of the data while reducing dimensionality. The pairwise scatter plots indicate that the relationships between features are largely maintained, although some transformations in the spread and alignment of the data points are visible, reflecting the effect of PCA in aligning the variance along the principal components. Additionally, the 3D plots demonstrate how the data is compactly represented in the new feature space while still capturing the essential patterns from the original dataset.

5 Comparison with Sklearn’s PCA

In this section, we compare the performance of our PCA implementation with sklearn’s PCA and Incremental PCA (IPCA) on the two datasets chosen (Hepatitis dataset and the Breast Cancer dataset). Figures 3 and 5 illustrate the reconstruction error and explained variance ratio for each method across varying numbers of components. The reconstruction error measures the loss of information during dimensionality reduction, with lower values indicating better preservation of the dataset’s original structure. On the other hand, the explained variance ratio reflects the proportion of the dataset’s variance captured by the selected components, where higher values indicate a more effective dimensionality reduction process.

5.1 Hepatitis

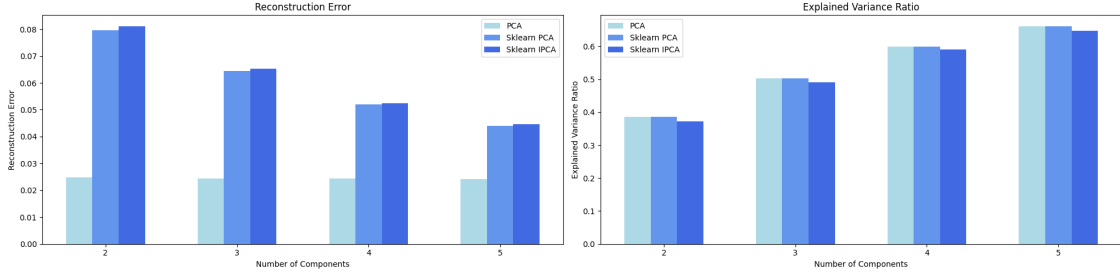


Figure 3: Reconstruction error and explained variance ratio comparison for PCA methods on the Hepatitis dataset.

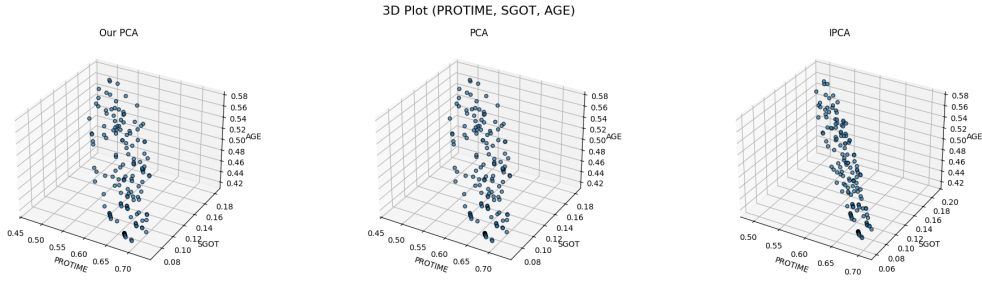


Figure 4: 3D comparison of our PCA, sklearn’s PCA, and Incremental PCA on the Hepatitis dataset features (*Protime*, *SGOT*, *Age*)

Figure 4 shows the 3D plots of the features *Protime*, *SGOT*, and *Age* after applying our PCA, sklearn’s PCA, and Incremental PCA (IPCA) on the Hepatitis dataset. These plots align with the findings from the earlier comparison of reconstruction error and explained variance ratio.

Our PCA achieves a consistently low reconstruction error (approximately 0.025, regardless of the number of components), while the reconstruction errors for sklearn’s PCA and IPCA are slightly higher, though still low. This indicates that all methods effectively preserve the original data structure, but our PCA demonstrates slightly better preservation.

The explained variance ratio for our PCA and sklearn’s PCA is identical, slightly outperforming the IPCA, which shows a marginally lower explained variance. As expected, the explained variance increases with the number of components for all methods, indicating that more components capture more variability in the dataset.

The similarity between the 3D plots of Figure 3 for our PCA and sklearn’s PCA confirms that both methods yield nearly identical results, preserving the data relationships in the reduced space. Meanwhile, the IPCA plot shows slight deviations, reflecting its lower explained variance ratio. This consistency further supports the accuracy of our PCA implementation in comparison to the sklearn standard.

5.2 Breast Cancer

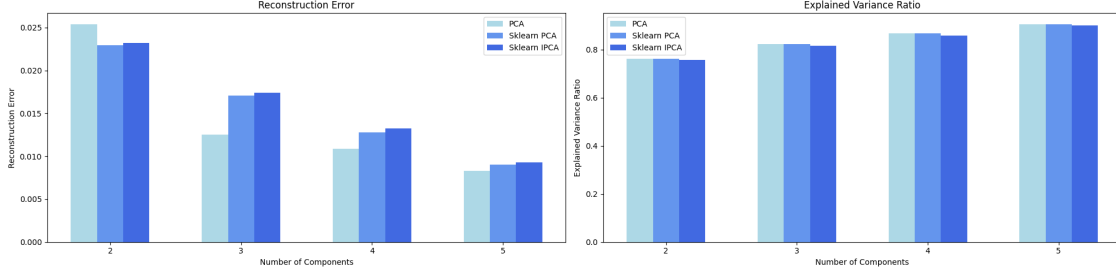


Figure 5: Reconstruction error and explained variance ratio comparison for PCA methods on the Breast Cancer dataset.

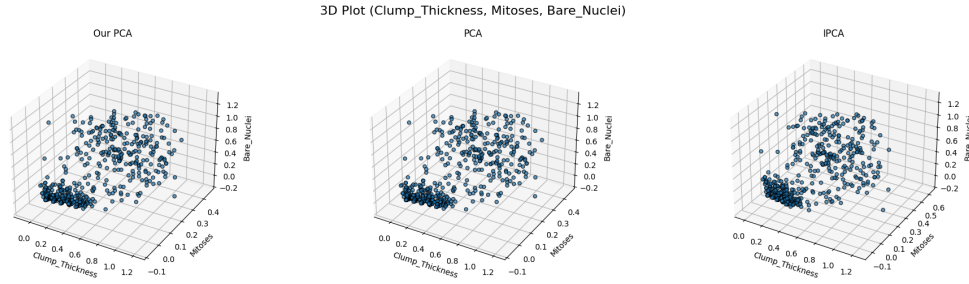


Figure 6: 3D comparison of our PCA, sklearn’s PCA, and Incremental PCA on the Breast Cancer dataset features (*Clump_Thickness*, *Mitoses*, *Bare_Nuclei*)

Figure 6 shows the 3D plots of the features *Clump_Thickness*, *Mitoses*, and *Bare_Nuclei* after applying our PCA, sklearn’s PCA, and Incremental PCA (IPCA) on the Breast Cancer dataset. The behavior observed here is consistent with that of the Hepatitis dataset, demonstrating that our PCA implementation performs similarly well across datasets of different types.

The reconstruction error for our PCA is generally low, comparable to the other methods, with a minor increase (approximately 0.002) when using two components. However, this increase is minimal and does not significantly affect the overall effectiveness of the method. Similar to the Hepatitis dataset, sklearn’s PCA and IPCA show slightly higher reconstruction errors than our PCA but still maintain low values, reflecting good preservation of the original structure.

The explained variance ratio follows the same pattern as in the Hepatitis dataset: our PCA and sklearn’s PCA yield identical explained variance ratios, both slightly higher than IPCA. As expected, the explained variance increases with the number of components, demonstrating effective dimensionality reduction for all three methods in this dataset too.

The 3D plots for our PCA and sklearn’s PCA are practically identical, confirming the consistency of their transformations. This similarity, combined with the low reconstruction error and high explained variance ratio, highlights that our PCA implementation functions effectively on both mixed (Hepatitis) and fully numeric (Breast Cancer) datasets, as expected.

6 Clustering with K-Means++ and Optics

This section evaluates the impact of dimensionality reduction on K-Means++ and OPTICS using Silhouette Score and Calinski-Harabasz Index as internal metrics, and Jaccard Index and Rand Index as external metrics. The analysis is performed on the Hepatitis and Breast Cancer datasets with three configurations: without PCA, with our PCA implementation, and with scikit-learn’s KernelPCA.

6.1 Hepatitis

Figure 7 reveals a consistent trend across both clustering techniques, K-Means++ and OPTICS. The application of our PCA and KernelPCA improves the internal clustering metrics, Silhouette Score and Calinski-Harabasz Index, for all tested numbers of components. This indicates that dimensionality reduction enhances the compactness and separation of the clusters as measured internally. However, this improvement in internal metrics comes at the expense of external metrics, Jaccard Index and Rand Index, which consistently deteriorate across all numbers of components.

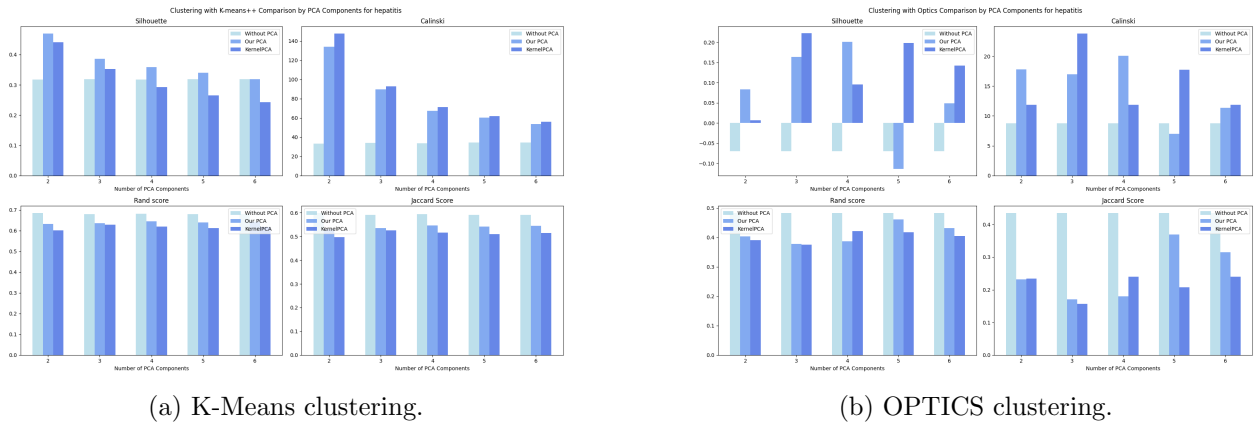


Figure 7: Internal (first row) and external (second row) analysis of K-Means++ and OPTICS for the Hepatitis dataset

For OPTICS, the degradation in external metrics is particularly pronounced, highlighting a significant misalignment between the clustering results and the ground truth labels when PCA or KernelPCA is applied. Regarding internal metrics, it’s also worth mentioning that, without PCA, the Silhouette score is negative or close to zero in most cases, implying poorly defined clusters when no dimensionality reduction is applied. However, no clear pattern emerges for OPTICS when varying the number of components, suggesting an inconsistent impact of dimensionality reduction even when dimensionality reduction is applied. For K-Means++, while PCA and KernelPCA improve internal metrics overall, these metrics tend to decrease as the number of components increases. Interestingly, the external metrics for K-Means++ remain relatively stable and show minimal variation with the number of components, indicating a steadier alignment with the ground truth labels.

6.2 Breast Cancer

For the Breast Cancer dataset, Figure 8 shows a similar trend in internal metrics, with both K-Means++ and OPTICS benefiting from the application of PCA and KernelPCA, particularly evident in the Silhouette score plot. In contrast to the Hepatitis dataset, the external metrics for

K-Means++ improve when either our PCA implementation or KernelPCA is applied, indicating better alignment with the ground truth labels.

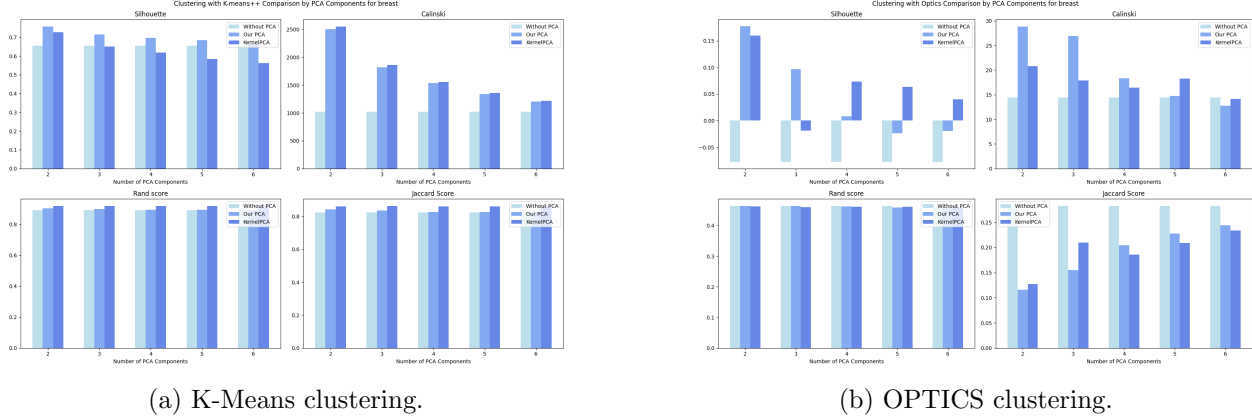


Figure 8: Internal (first row) and external (second row) analysis of K-Means++ and OPTICS for the Breast Cancer dataset

As the number of components increases, both clustering algorithms exhibit a decline in internal metrics, with the performance worsening as more components are considered. However, the external metrics remain largely stable, with the exception of the Jaccard Index for OPTICS, which shows an improvement with higher numbers of components. This suggests that while the internal clustering quality decreases with more components, certain external alignments, such as the Jaccard score for OPTICS, may still benefit from dimensionality reduction.

For both the Hepatitis and Breast Cancer datasets, no clear pattern emerges between our PCA implementation and KernelPCA from scikit-learn. Depending on the dataset and clustering algorithm, one method tends to outperform the other, highlighting the variability of PCA-based approaches in different contexts.

7 Visualization

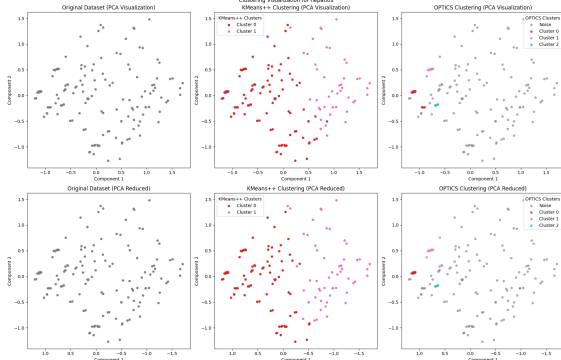
In this section, we conduct two experiments to compare clustering visualizations from OPTICS and K-Means++. In the first experiment, we apply PCA for both dimensionality reduction and visualization of each dataset. In the second experiment, we use UMAP for visualization while applying PCA for dimensionality reduction.

7.1 PCA

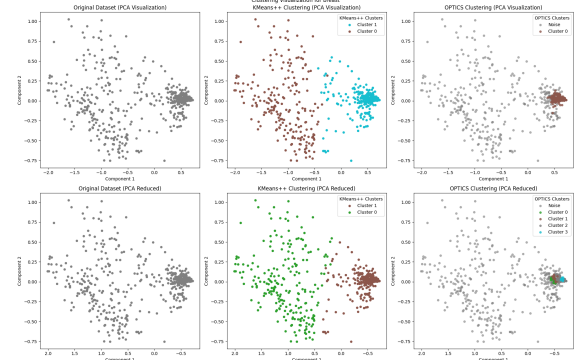
The results of PCA applied for visualization and dimensionality reduction can be seen in Figure 9a for the Hepatitis dataset and in Figure 9b for the Breast Cancer dataset. In both figures, the first row represents PCA directly applied to the dataset with two components ($n=2$). The second row shows the results after first applying PCA to reduce the dimensionality, retaining 80% of the variance, and then visualizing the reduced data with two components ($n=2$).

For the Hepatitis dataset, there is no significant visual difference between the two methods, aside from a few points on the edge of the clusters. However, for the Breast Cancer dataset, a clearer distinction is observed, particularly in the OPTICS clustering. Initially, there was one large cluster,

but after applying PCA for dimensionality reduction, the cluster has been segmented into smaller clusters with distinct sizes, along with several noise points, highlighting the impact of dimensionality reduction on the clustering structure.



(a) PCA visualization of the Hepatitis dataset.

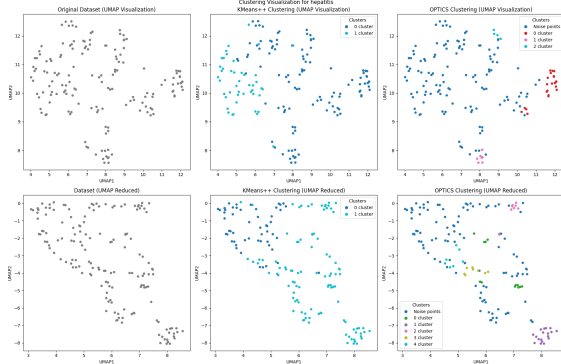


(b) PCA visualization of the Breast dataset.

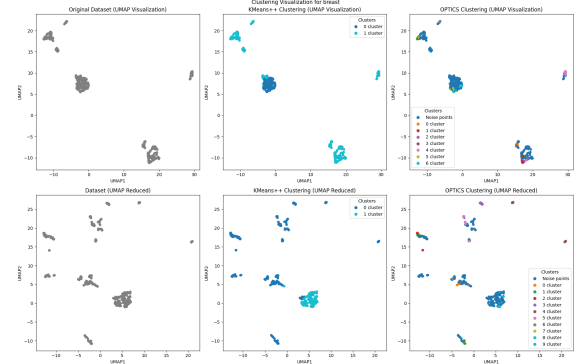
Figure 9: PCA visualization for the Hepatitis and Breast Cancer datasets

7.2 UMAP

Figure 10a shows the clustering results for the Hepatitis dataset, while Figure 10b presents the results for the Breast Cancer dataset. In both figures, the first row illustrates the clustering results visualized with UMAP. The second row displays the outcomes after applying PCA for dimensionality reduction (retaining 80% of the variance), followed by UMAP visualization with two components ($n=2$).



(a) UMAP visualization of the Hepatitis dataset.



(b) UMAP visualization of the Breast dataset.

Figure 10: UMAP visualization for the Hepatitis and Breast Cancer datasets

For both datasets, significant differences are observed compared to the previous PCA approach. UMAP visualization reveals the clusters formed by both K-Means++ and OPTICS more clearly. Two key insights emerge. Firstly, dimensionality reduction has a more noticeable effect, with the representation of points looking markedly different from the original dataset. Second, for both datasets, while the clusters from both clustering techniques are visible, no clear pattern emerges, as the clusters appear in distinct locations. The only relevant insight is that optics clusters are smaller and located in areas with high density of points.

8 Conclusions

In this study, we applied dimensionality reduction, clustering, and visualization techniques to the Hepatitis dataset (mixed data types) and the Breast Cancer dataset (numerical). Dimensionality reduction using PCA and UMAP was effective in simplifying the data while retaining most of its variance. PCA provided an interpretable linear reduction and performed competitively compared to sklearn’s PCA and IPCA, showing slight advantages in handling datasets with mixed features. On the other hand, UMAP offered clearer visual cluster separations and highlighted high-density areas, making it particularly useful for complex data.

For clustering, K-Means++ performed well, identifying consistent and distinct clusters across both datasets, even after dimensionality reduction. In contrast, OPTICS struggled to produce meaningful clusters, likely due to its reliance on density-based assumptions that did not align well with the structure of the data. A key observation was the impact of dimensionality reduction on clustering metrics. While internal metrics improved, external metrics showed some decline. This trade-off highlights the balance between internal clustering structure and alignment with true labels.

Visualization with PCA and UMAP showed differences: UMAP provided a more evident separation of clusters and revealed smaller, denser clusters formed by OPTICS, though these were less coherent. Additionally, dimensionality reduction significantly reduced computational time, especially with UMAP, without compromising clustering results.

Overall, while dimensionality reduction and K-Means++ proved effective, careful consideration is essential when selecting dimensionality reduction methods. The challenges faced by OPTICS with these datasets underscore the importance of aligning methods with the data’s characteristics.

References

- [1] Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 1027-1035.
- [2] Celebi, M. E., Kingravi, H. A., & Vela, P. A. (2012). *A comparative study of efficient initialization methods for the K-means clustering algorithm*.
- [3] Maria Salamó Llorente (2024). *Theory 10. Visualization*. Universitat de Barcelona.